

PROJECT AND TEAM INFORMATION

Project Title

Food Surplus Management and Redistribution System

Student / Team Information

<i>Team Name:</i> <i>Team #</i>	<i>FoodRescuers</i> <i>DSCPP-III-2026-T074</i>
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Every day, tonnes of usable food are left unused and wasted in weddings, events, hostel messes, college canteens, hotels and various places, while some people are not even able to receive a proper and nutritious diet. Food wastage is not merely a small issue but a significant problem that the society faces. Hence, surplus food should be properly managed and redistributed to people in need, orphanages, NGOs and even to animal shelters, before it gets expired or spoiled.

The problem is not simply the availability of food but the lack of timely coordination between donors and suitable recipients. Therefore, it is necessary that the right quantity and type of food reaches the right place at the right time. This project aims to develop a system to solve this problem of food wastage as well as the problem of food insecurity amongst the people who cannot afford adequate amount of food.

State of the Art / Current solution (1 pt)

Currently, surplus food is managed and redistributed through social media platforms or through regular calls and texts between the donor and the recipient. This makes it difficult to check the quality, quantity and type of food being redistributed. This also leads to confusion about the location and time at which the food will be delivered making it unfit for consumption if the food has short expiry window.

Manual work is prone to errors and negligence which means that insufficient importance is given to important factors like food expiry window, allergen information, dietary requirements, urgency, location and various other factors. This project not only manages the significant problem of food wastage but also deploys a proper system to consider all the necessary factors for collecting and redistributing surplus food.

Project Goals and Milestones (2 pts)

The overall goal of this project is to develop a food surplus management and redistribution system that collects surplus usable food from donors and redistribute it to the suitable recipient before it expires.

Phase 1 Goals- *The main goals of Phase 1 are listed below:-*

S No.	Features	Status
1.	System Architecture and User Workflow Design	Completed on 24 August
2.	Integrated Web-Based Front-End Development	Completed on 30 August
3.	C++ Object-Oriented System Foundation	To be completed by 5 September
4.	Primary Data Structure Implementation in C++	To be completed by 12 September
5.	Core Backend Logic and Algorithmic Processing	To be completed by 20 September
6.	Integrated End-to-End System Workflow	To be completed by 25 September

Initial Milestones:-

- *Requirement and System Design: Finalize requirements, user roles, system architecture, OOP class structure, data-structure plan and UI wireframes.*
- *Frontend Development: Develop the basic donor and recipient oriented interface, including dashboard, food registration form, food listings and food details.*
- *30% Backend Development: Implement the Food class, basic User/Recipient classes, food validation, Linked List inventory, Min-Heap urgency management, Hash Table lookup/allergen mapping, Queue foundation and C++ file persistence.*
- *UI and Logic Integration: Connect the Phase 1 frontend workflow with the implemented backend/core logic and demonstrate the registration and food-management flow.*
- *Testing: Test valid and invalid food entries, expiry conditions, allergen conflicts, empty data structures, duplicate IDs and basic recipient matching.*

The future milestones include development of proper food surplus management and redistribution system with advanced technologies like delivery tracking and management, sponsorship and AI/ML optimized system.

Project Approach (3 pts)

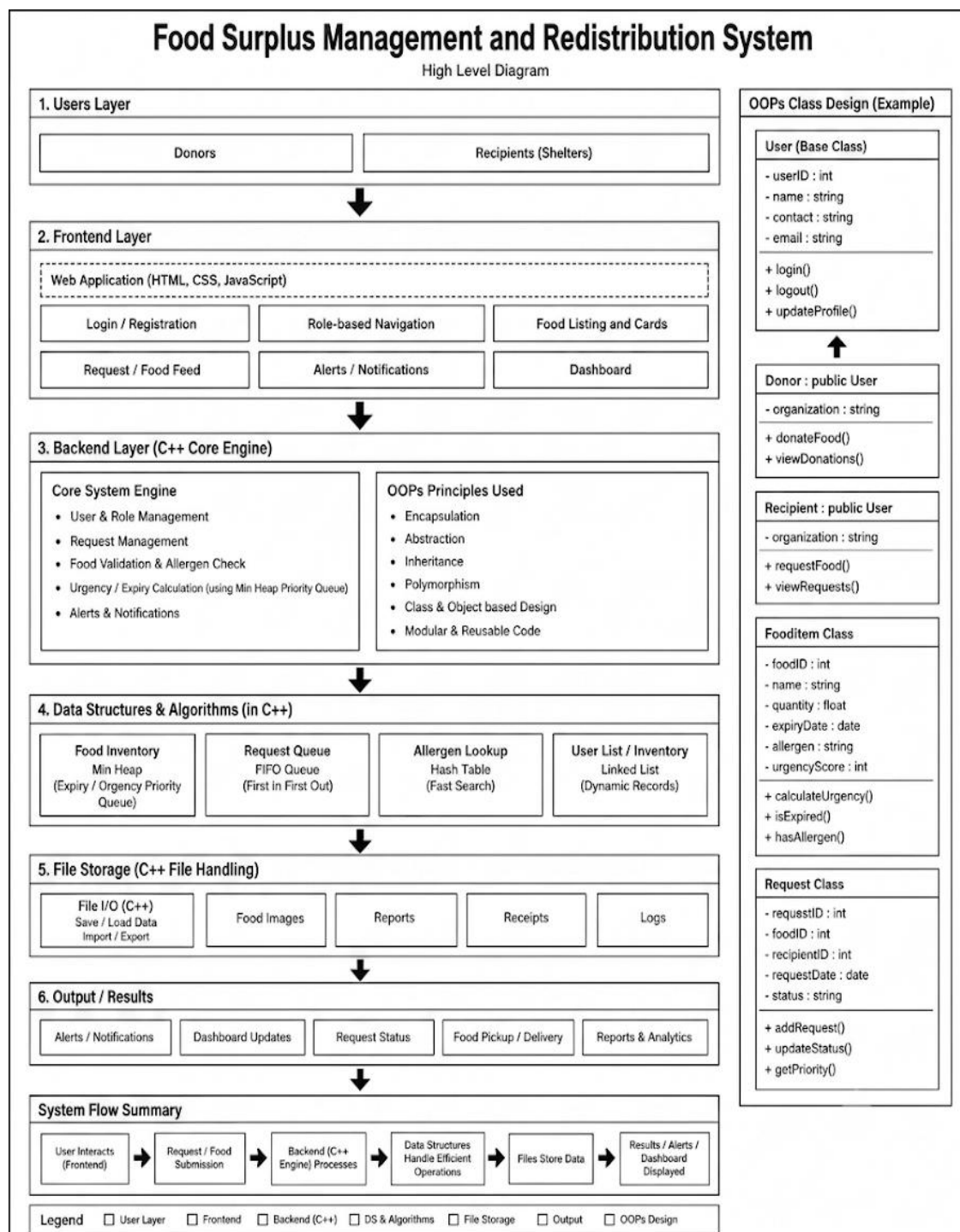
- *This project's core backend will be based on C++ Object Oriented Programming and Data Structures and frontend will be based on HTML, CSS, JavaScript with UI integration.*
- *The C++ backend will use Object Oriented Programming concepts such as Encapsulation for privacy, Inheritance, abstraction and polymorphism. The main entities are user, donor, recipient, food, request, pickup and drop.*
- *The integration of data structures in C++ is an important part of the system backend. The various used data structures are linked list for maintaining dynamic food inventory, a Min-Heap for prioritizing food according to expiry window, a hash table for fast Food ID and ingredient-to-allergen lookup, a queue for managing recipient requests as per first come and first serve basis.*
- *Algorithms that will be used include Linear/Binary search, Bubble/Selection Sort, Hash Indexing, FIFO mechanism, Heapify operations, weighted multi-criteria matching and file handling.*
- *C++ File I/O will be used to store and manage basic system records.*
- *The frontend will provide the user interface through forms, dashboard, food cards and basic request/matching views.*

System Architecture (High Level Diagram)(2 pts)

The system consists of five major layers:

1. Users:- Donors, Recipients/NGOs/Shelters
2. Frontend is developed using HTML, CSS, JavaScript. The frontend provides: Login/registration, role-based navigation, donor food registration, food listing and food cards, recipient food feed, request interface, alerts and basic dashboard
3. C++ core processing backend contains modules for: User management, food management, food validation, urgency/expiry calculation, allergen checking, basic recipient matching, request management.
4. Data Structures:- The backend actively uses linked list for food inventory, min-heap for expiry/urgency prioritization, hash table for fast ID and allergen lookup, FIFO Queue for request management.
5. The project uses C++ File I/O for saving/loading data.

High-Level Flow Diagram:-



Project Outcome / Deliverables (1 pts)

At the completion of Phase 1, the following outcomes are expected:

- *A functional frontend using HTML, CSS and JavaScript.*
- *Role-oriented interfaces for donors and recipients.*
- *A donor food-registration form containing food name, quantity, preparation time, best-before time, location, ingredients, diet type and packaging information.*
- *Food listing and food-detail interfaces. Basic food validation and expiry/urgency calculation.*
- *C++ Food and user-related classes forming the foundation of the backend.*
- *Linked List implementation for food inventory. Min-Heap implementation for urgent food prioritization. Hash Table implementation for Food ID and ingredient/allergen lookup. FIFO Queue foundation for recipient requests.*
- *Basic C++ File I/O for persistence.*
- *Basic rule-based recipient matching.*
- *A working Phase 1 demonstration showing the flow from food registration to basic matching/display.*
- *Phase 1 source code, documentation and test cases.*

Assumptions

The following assumptions were considered for solving the problem:-

- *Donors provide accurate information about food quantity, ingredients, preparation time and expiry window.*
- *Recipients provide accurate dietary restrictions, allergies, food types and quantity.*
- *Food is assumed to be eligible for redistribution only after the project's defined safety checks.*
- *C++ File I/O is sufficient for fulfilling Phase 1 requirements.*
- *Recipients will pick the food themselves from the pickup location.*

References

- GeeksforGeeks (Data Structures and Algorithms in C++):- <https://www.geeksforgeeks.org/>
- MDN Web Docs (HTML, CSS and JavaScript documentation):- <https://developer.mozilla.org/>
- cppreference (C++ Language and Standard Library Reference):- <https://en.cppreference.com/>
- Food and Agriculture Organization of the United Nations (FAO) (Food loss and food waste resources):- <https://www.fao.org/>